

LESSON GUIDE

Technology Plus

Unit 7: Security

6.1.5 – AAA Authentication

Lesson Overview:

In this lesson, students will explore the different factors of authentication and how they work to confirm a user's identity.

Students will:

- Explain what authentication is and why it is important for digital security.
- Differentiate between authentication methods.
- Evaluate which authentication method would be the most effective in different scenarios.

Guiding Question: How can you prove your identity online?

Suggested Grade Levels: 8-12

Technology Needed: No

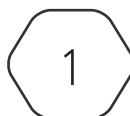
Approximate Time Required: 40-55 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 6.1 - Explain fundamental security concepts and frameworks.
 - Authentication, authorization, accounting, and non-repudiation concepts
 - **Authentication**
 - **Single factor**
 - **Multifactor**
 - **Single sign-on**
 - Authorization
 - Permissions
 - Administrator vs. user accounts
 - Least privilege model
 - Accounting
 - Logs
 - Location tracking
 - Web browser history

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Lesson Background

Background Information:

In today's digital world, protecting information and verifying identity are essential parts of cybersecurity. The AAA triad—Authentication, Authorization, and Accounting— provides a framework for managing access and tracking user activity within systems. An additional concept, non-repudiation, ensures that users cannot deny their actions once they have been verified and recorded.

This lesson focuses on the first piece of the triad: Authentication. Authentication is the process of confirming that someone is who they claim to be before granting access to a system or resource. Students encounter authentication every day when they log into phones, email accounts, or social media. Understanding authentication helps them recognize the difference between single-factor authentication (SFA), which uses only one method like a password, and multi-factor authentication (MFA), which combines two or more methods such as a password, a phone code, or a fingerprint. Students will also explore Single Sign-On (SSO), which allows one login to grant access to multiple systems or applications.

By studying these methods, students will not only learn how systems keep users secure but also why stronger authentication methods are critical in protecting data from unauthorized access. This background builds a foundation for understanding broader security frameworks and prepares students to evaluate the effectiveness of different authentication strategies in both personal and professional settings.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 6.1.5 – AAA Authentication • Activity Slides 6.1.5 – Access Granted
Materials per Student	• Student Notes 6.1.5 – AAA Authentication • Student Handout 6.1.5 – AAA Authentication • Assessment 6.1.5 – AAA Authentication (optional)

Lesson Background

Cyber Terms:

Term	Definition
AAA Triad	a cybersecurity framework that secures access to systems and data by making sure users are who they say they are (Authentication), can only do what they're allowed to do (Authorization), and that their actions are recorded (Accounting)
authentication	the process of verifying who a user is before granting access to an account: something you know, something you have, or something you are
single-factor authentication (SFA)	relies on one form of verification before granting access
multi-factor authentication (MFA)	relies on two or more different types of verification before granting access
single sign-on (SSO)	allows users to log in once and gain access to multiple systems or applications without re-entering credentials

6.1.5 – AAA Authentication

Guiding Question: How can you prove your identity online?

Lesson Launch (5 minutes)

- You pick up your phone and try to unlock it. You enter your password, but then your phone also asks for a fingerprint.
- You attempt to log into your favorite social media app, when you type in your password it sends a code to your phone.
- You log into a school computer with your password and can open your email, your files, and your assignments without logging in again.
- All of these actions are part of a process called **authentication**; it is how systems make sure that you are really you.
- Today you will explore different factors of authentication and how they work.

Authentic Authentication (20-25 minutes)

- Distribute the Student Handout 6.1.5 – AAA Authentication and have students fill in the answers during the lesson.
- Introduce the **AAA Triad** and its relationship with non-repudiation
 - Authentication: Ensures that the right people are logging in
 - Authorization: Ensures users can only do things they are allowed to
 - Accounting: Ensures a user's actions are recorded for accountability
 - Together, these concepts make sure that digital systems know who is accessing them, control what they can do, track their actions, and ensure those actions cannot be denied later.
 - In practice, the AAA Triad is the foundation of many security systems, such as logging into school accounts, accessing files on a network, or managing user permissions in an organization.
 - It helps organizations maintain security, protect sensitive information, and track activity for auditing or troubleshooting.
- Discuss **authentication**, its definition, its features, and its factors.
 - Authentication protects your accounts and data from unauthorized access, prevents people from pretending to be you, and ensures that only verified users can use apps, files, or services.
 - Authentication:
 - Protects Accounts and Data: Stops unauthorized people from accessing personal or sensitive information

- Prevents Identity Fraud: Makes it harder for someone to impersonate you online
- Supports Accountability: Helps track who is doing what in a system
- Enables Secure Access: Ensures only verified users can use certain apps, files, or services
- Factors include something you know, something you have, and something you are, to prove your identity and gain access to an account.
- Introduce **single-factor authentication (SFA)** as a method of authentication and some examples.
 - Although easy and quick, this is not the most secure form of authentication. Passwords or PINs are vulnerable to attacks, like brute force or phishing, and can be easily guessed or stolen.
 - Even though face scan and fingerprint are advanced technology, it can still be single-factor authentication if it is not combined with anything else like a password or a code.
 - SFA can be any factor of authentication, as long as it grants access after the use of only one.
- Introduce **multi-factor authentication (MFA)** as a method of authentication and share two examples.
 - Verification types include multiple factors of authentication, making it harder for unauthorized users to access the account.
 - MFA can include any combination of factors.
 - You may also hear of two-factor authentication (2FA), it includes exactly two factors to verify identity. All 2FA is MFA, but not all MFA is 2FA.
- Introduce **single sign-on (SSO)** as a method of authentication and share an example.
 - If an attacker gets into your SSO account, they can access everything linked to it.
 - SSO is one login with access to many systems.

Activity: Access Granted (10-15 minutes)

- Display Activity Slides 6.1.5 – Access Granted, have students record their answers in the handout.
- Decide if students will work independently or in small groups.
- Read each scenario aloud (or allow students/groups to read independently).
- Students will determine what if the authentication method was single-factor, multi-factor, or single sign-on.
- Allow time for students to write their answers.

Scenarios and Answers

- Evan logs into his social media account by typing only his username and password. Once he is in, he can start posting.

Single-Factor Authentication (SFA)

- After typing her email password, Sophia receives a text message with a code. She must enter the code to finish logging into her account.

Multi-Factor Authentication (MFA)

- Malik signs into his school's cloud account one time. After that, he can open documents, email, and an online workspace without logging in again.

Single Sign-On (SSO)

- Ryleigh logs into a gaming platform using just her fingerprint on the console.

Single-Factor Authentication (SFA)

- Mia signs in to her online banking account with a password. Then she confirms her identity with a fingerprint scan on her phone.

Multi-Factor Authentication (MFA)

- After entering his password on the school computer, Aiden is automatically logged into his email, online classes, and cloud storage.

Single Sign-On (SSO)

- Harper unlocks her tablet with a six-digit PIN. Once the code is entered her device is fully accessible.

Single-Factor Authentication (SFA)

- Before logging into his workplace system, Noah enters a password, uses a smart card, and scans his fingerprint to gain access.

Multi-Factor Authentication (MFA)

- Ava signs into her account once. From there she can access multiple systems without entering her credentials again.

Single Sign-On (SSO)

- Leo logs into his school library website with a password. He then enters a code sent to his phone before accessing the system.

Multi-Factor Authentication (MFA)

Putting It All Together (5-10 minutes)

Discussion Questions

- Review what students learned today by discussing the questions below as a class.
- What are the three main factors of authentication?

The three main factors of authentication are something you know, something you have, and something you are.

- What is the difference between SFA, MFA, and SSO?

Single-factor authentication (SFA) relies on one form of verification, multi-factor authentication (MFA) relies on two or more different types of verification, and single sign-on (SSO) allows users to log in once and gain access to multiple systems or applications without re-entering credentials.

- Why is MFA considered stronger than SFA?

Attackers would need to break more than one type of factor at the same time.

- Think about your own digital life: Where do you already use SFA, MFA, and SSO? Which one do you think protects you the best, and why?

Ex: SFA = Netflix password; MFA = bank login with code; SSO = school Google account. MFA protects best because it requires multiple proofs of identity.